

Day 2 · Taxonomy and Community Ecology

Marine eDNA Metabarcoding Analysis using R

Jeju ARMS Case Study

Song, Chi-une

2026.09.01

WORKSHOP OVERVIEW

Overview — workshop workflow

Day 1 generated ASVs; Day 2 connects them to ecology.

01

Sampling & DNA Preparation

Collect and prepare the biological material

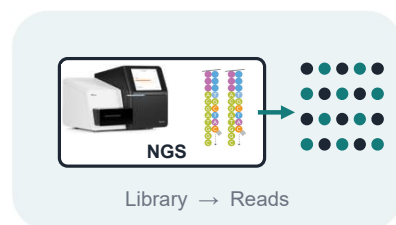


- 1 Marine sample collection
- 2 DNA extraction
- 3 COI barcode amplification

02

NGS Sequencing

Convert amplified DNA into sequencing reads



- 1 Library preparation
- 2 Illumina sequencing
- 3 Raw sequencing output

03

Bioinformatics Processing

Turn raw reads into reliable sequence variants



- 1 FASTQ quality control
- 2 Filtering & denoising
- 3 ASV inference with DADA2

04

Ecological Analysis

Connect sequence variants to community ecology



- 1 Taxonomic assignment
- 2 Alpha & beta diversity
- 3 Ecological interpretation

DAY 1 · Raw reads → ASVs

DAY 2 · ASVs → Ecology

Today's four questions

ASV sequences are the starting point, not the final answer.

Taxonomy & filtering → α -diversity → β -diversity → Indicator taxa

1. What organisms match our ASVs, and which should we analyze?

→ BLAST search + Taxonomic & ecological filtering

2. How diverse is each community?

→ Rarefaction + α -diversity

Richness · Shannon · Simpson

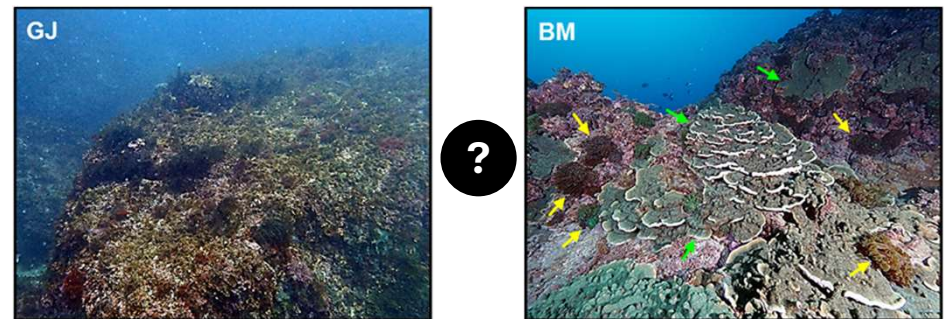
3. How do the two communities differ?

→ β -diversity

Bray-Curtis · nMDS · ANOSIM

4. Which taxa characterize each site?

→ Indicator Value (IndVal)



Gangjeong vs Bomok

Two Jeju habitats

Gangjeong and Bomok represent contrasting benthic settings.

GJ

Gangjeong

Macroalgae-dominated habitat

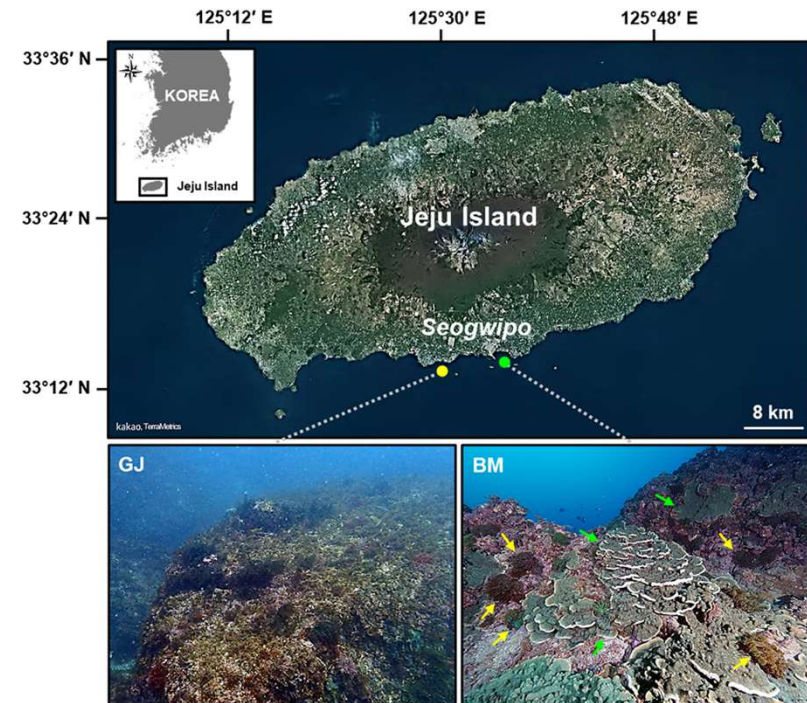
- Turf-forming algae
- Gangjeong1–Gangjeong4

BM

Bomok

Coral-dominated habitat

- Scleractinian corals
- Bomok1–Bomok4



Study sites on the southern coast of Jeju Island

Lee et al. (2024)

KEY POINT • Eight teaching profiles are compared as two groups of four.

What Day 1 produced

Day 1 produced an ASV abundance table and representative sequences.

ASV TABLE

- Rows: unique ASVs
- Columns: samples
- Cells: read counts

ASV_ID	GJ1	GJ2	GJ3	GJ4	BM1	BM2	BM3	BM4
ASV25	328	317	228	200	34	24	26	36
ASV26	371	888	356	384	191	57	36	89
ASV28	0	1010	136	101	2734	12201	263	3129
ASV29	420	361	206	416	0	25	25	44
ASV30	91	24	40	22	0	29	10	17
ASV31	44	159	227	270	804	636	1873	883
...								

REPRESENTATIVE SEQUENCES

- One DNA sequence per ASV
- No organism name yet

>ASV31

```
AAC TTTATATTTAATTTTTGGG CCTGATCAGCTATAGTAGGAACAGCCTTAAGAATGTT
AATTCGAGCTGAATTAGGTCAACCAGGAAGTTTAATTGGTGATGATCAAATTTATAATGT
TATTGTTACAGCTCAGCTTTTATTATAATTTCTTTATAGTAATACCTATTATAATTGG
GGGATTTGGAAATTGACTTCTTCCTTTAATACTTGGTGCCCTGATATGGCATTCCCTCG
ATTAATAATATGAGATTTTGACTTCTTCCCCCAGC
```

>ASV452

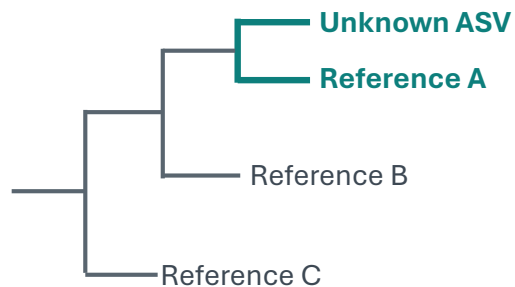
```
AACCTTATATTTAATTTTTGGCGCGTTTTCTGGAGTTTTAGGTGCGTGATTTCTATGTT
AATTCGTATGGAATTAGCTCAACCAGGCAATCACTTACTACTTGGTAATCATCAAGTTTA
CAATGTACTTATAACAGCTCATGCTTTTTTAATGATATTTTTTTAGTTATGCCAGTAAT
GATAGGAGGTTTTGGTAATTGATTAGTTCCTATTATGATTGGTAGCCCTGATATGGCTTT
TCCTCGTTTAAATAATATTAGTTTTTGATTATTACC
```

...

Related organisms have similar barcode sequences

Closely related organisms generally have more similar sequences in the same barcode region.

Evolutionary relationship



Closer relationship

→ fewer sequence differences

Barcode comparisons

Unknown ASV	A T G G C A G G T A C C T G A T C A G T	query
Reference A	A T G G C A G G T A C C T G A T C A G <u>C</u>	high similarity
Reference B	A T G <u>A</u> C A G G T <u>T</u> C C T G A T C A G <u>C</u>	lower similarity
Reference C	A <u>C</u> G <u>A</u> T <u>C</u> G <u>A</u> T <u>C</u> <u>C</u> G <u>T</u> A A T C G A <u>C</u>	low similarity

More similar barcode sequence → stronger evidence for taxonomic similarity

Reference databases

Taxonomic assignment compares an unknown ASV with identified reference sequences.

QUERY

Unknown ASV (ASV452)

>ASV452

```
AACCTTATATTTAATTTTTGGCGC
GTTTTCTGGAGTTTTAGGTGCGTG
TATTTCTATGTTAATTCGTATGGA
ATTAGCTCAACCAGGCAATCACTT
ACTACTTGGTAATCATCAAGTTTA
CAATGTACTTATAACAGCTCATGC
TTTTTTAATGATATTTTTTTTAGT
TATGCCAGTAATGATAGGAGGTTT
TGGTAATTGATTAGTTCCTATTAT
GATTGGTAGCCCTGATATGGCTTT
TCCTCGTTTTAAATAATATTAGTTT
TTGATTATTACC
```

Compare

REFERENCE DATABASE

Mesophyllum sp. 4 CP-2018 voucher VPF00480 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial

GenBank: MF133365.1

[FASTA](#) [Graphics](#)

Go to: ☐

LOCUS MF133365 664 bp DNA linear PLN 21-APR-2018

DEFINITION Mesophyllum sp. 4 CP-2018 voucher VPF00480 cytochrome oxidase subunit 1 (COI) gene, partial cds; mitochondrial.

ACCESSION MF133365

VERSION MF133365.1

KEYWORDS

SOURCE mitochondrion Mesophyllum sp. 4 CP-2018

ORGANISM [Mesophyllum sp. 4 CP-2018](#)

Eukaryota; Rhodophyta; Florideophyceae; Corallinophycidae; Hapalidiales; Hapalidiaceae; Mesophylloideae; Mesophyllum.

```
1 aaccttatat ttaattttt ggcgttttc tggagtttta ggtgcgtgta tttctatgtt
61 aattcgtatg gaattagctc aaccaggcaa tcacctacta cttggtaatc atcaagttaa
121 taatgtactt ataacagctc atgctttttt aatgatat tt ttttagtta tgccagtaat
181 gataggaggt ttggttaatt gattagtcc tattatgatt gtagtcccg atatggcctt
241 tctcgttta aataatatta gtttttgatt attaccgcct tcattatgct ttttattagc
301 ttcttctatt gttgaagtag gcgcaggtag aggatgaaca gtgtaccac ccttaagttc
361 tatccaaagt cattctggag catcagtga cttagcaatt ttagtttac atttagcagg
421 agcatcttca atacttggtg ctattaattt tatatcaact attttaaata tgcgtaatcc
481 tggtaaaagt atgtatagaa ttccattatt tgtttgatca atttttgta ctgcattttt
541 acttttattg gcagtagctg ttttagctgg tgcattaca atgtatttaa ctgatagaaa
601 ttttaatact tgcgtttttg atctgcagg agggagtgat ccaattttg atcaacattt
661 attt
```

Taxonomic classification

DNA barcode sequence



[Mesophyllum sp. 4 CP-2018 voucher VPF00480 cytochrome oxidase subunit - Nucleotide - NCBI](#)

KEY POINT • Reference databases link known DNA sequences to taxonomic names.

What BLAST does

BLAST rapidly searches a large database for sequences similar to our query.

1 WHAT IS BLAST?

- **B**asic **L**ocal **A**lignment **S**earch **T**ool
- Finds locally similar sequence regions in a large database.

2 HOW DOES IT WORK?

Query sequence → find short matching regions → extend promising matches → rank the strongest hits.

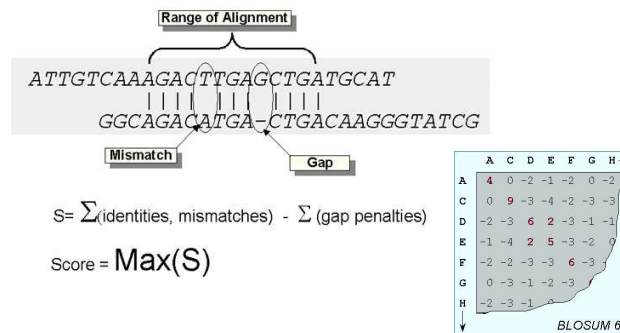
3 WHY IS IT USEFUL?

- Fast database search
- Finds similar regions without requiring full-length alignment
- Reports quantitative match metrics

SEARCH PRINCIPLE

BLAST is a local alignment search

- It first finds promising short sequence matches and then extends them to identify stronger local alignments.



[Download](#)
[GenBank Graphics](#)

Balanus prugnus voucher CR7 cytochrome c oxidase subunit I (COX1) gene, partial cds;
 Sequence ID: [PT718007.1](#) Length: 658 Number of Matches: 1

[Range 1: 1 to 276](#)
[GenBank](#)
[Graphics](#)
[▼ Next Match](#)
[▲ Previous Match](#)

Score	Expect	Identities	Gaps	Strand
150 bits(276)	4e-140	276/276(100%)	0/276(0%)	Plus/Plus
Query 1	AACCTTTATATTTAAATTTTGGGGCGTGATCAGCTATAGTAGGAAACAGCCTTAAAGATGTT	60		
Sbjct 1	AACCTTTATATTTAAATTTTGGGGCGTGATCAGCTATAGTAGGAAACAGCCTTAAAGATGTT	60		
Query 61	AATTCGAGCTGAATTAAGTCAACCGAAGGTTAAATGGTGATGCAAAATATATAAATG	120		
Sbjct 61	AATTCGAGCTGAATTAAGTCAACCGAAGGTTAAATGGTGATGCAAAATATATAAATG	120		
Query 121	TATGTGTACAGCTCAGCGTTTATATAAATTCCTTTATAGTAAATCCTATTATAATGG	180		
Sbjct 121	TATGTGTACAGCTCAGCGTTTATATAAATTCCTTTATAGTAAATCCTATTATAATGG	180		
Query 181	GGGATTGGGAAATGACCTTCTTCCTTTAAATCTTGGTGGCCCGTAGATGGGATTCCTCG	240		
Sbjct 181	GGGATTGGGAAATGACCTTCTTCCTTTAAATCTTGGTGGCCCGTAGATGGGATTCCTCG	240		
Query 241	ATTAATAAATATAGAGATTGGACTTCTCCCCCAGC 276			
Sbjct 241	ATTAATAAATATAGAGATTGGACTTCTCCCCCAGC 276			

Reading a BLAST result

Use several metrics to evaluate whether a match is informative.

Two values determine acceptance in this workshop dataset: **percent identity** and **query coverage**.

Key BLAST metrics

Percent identity – matching base within the aligned region

Query coverage – how much of the ASV sequence is aligned

Alignment length – number of bases compared

E-value – expected matches occurring by chance

Bit score – Overall strength of the alignment

Key BLAST metrics

Sequences producing significant alignments

Download Select columns Show 100 ?

☒ select all 100 sequences selected

GenBank Graphics Distance tree of results MSA Viewer

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Balanus trigonus voucher CR7 cytochrome c oxidase subunit I (COX1) gene, partial cds; mitochond...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	PP718007.1
☒	Balanus trigonus voucher cr10 cytochrome c oxidase subunit I (COX1) gene, partial cds; mitochond...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	PP718004.1
☒	Balanus trigonus mitochondrion, complete genome	Balanus trigonus	510	510	100%	4e-140	100.00%	15560	MW646099.1
☒	Balanus trigonus voucher UF-Invertebrate Zoology:45507-Arthropoda cytochrome c oxidase subunit...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	MW277822.1
☒	Balanus trigonus voucher CR9 cytochrome c oxidase subunit I (COX1) gene, partial cds; mitochond...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	PP718008.1
☒	Balanus trigonus voucher BHKG-1689 cytochrome oxidase subunit 1 (COI) gene, partial cds; mito...	Balanus trigonus	510	510	100%	4e-140	100.00%	658	PP652141.1

coverage

identity

KEY POINT •

High identity over a very short alignment can still be misleading.

Why identity alone is not enough

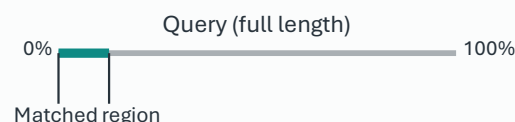
Interpret percent identity together with query coverage.

Two example matches

Example A

100% identity

10% query coverage



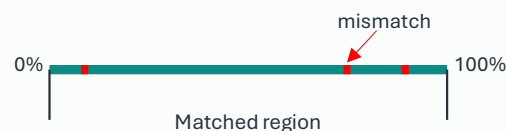
Only a short region matched

→ Weak evidence

Example B

97% identity

100% query coverage



Nearly the full ASV matched

→ Stronger evidence

- A perfect match over a tiny region can be less informative than a near-perfect match over the full sequence.

The acceptance rule



Query coverage $\geq 85\%$



Percent identity threshold depends on **taxonomic level**

Species	$\geq 98\%$
Genus	$\geq 95\%$
Family	$\geq 90\%$
Order	$\geq 85\%$

Why does identity start at 85%?

Because this workshop retains assignments down to the **order level**.



These thresholds are practical criteria for this analysis. They are not universal cut-offs for all taxa or barcode markers.

KEY POINT · Reliable assignment requires both sufficient coverage and an appropriate identity threshold.

What happened to 1,052 ASVs?

Let's inspect the taxonomy table and summarize the BLAST outcomes.

PRACTICAL

Inspect the taxonomy table

Load the file, preview its columns, and summarize BLAST status.

```
# Load the taxonomy table
taxonomy <- read.delim(
  "data/Day2/taxonomy_final_ASV.tsv",
  stringsAsFactors = FALSE)

# Preview the table
head(taxonomy)
names(taxonomy)

# How many ASVs?
nrow(taxonomy)

# Summary of BLAST assignment
table(taxonomy$BLAST_status)
```

BLAST assignment summary

One row per ASV • 1,052 total ASVs

322 **Accepted**
Passed the BLAST criteria

692 **Failed cutoff**
Did not meet identity / coverage criteria

38 **No hit**
No reference match was recovered

1,052 ASVs → **BLAST filtering** → 322 accepted

Why BLAST is precomputed

The time is reserved for interpreting results in R.

RUNNING LOCAL BLAST REQUIRES

1 BLAST+ installation

Install and configure the BLAST command-line tools

2 Reference database

Download sequence records for the target marker

3 Database indexing

Format the database before searching

4 System setup

Paths and commands differ across operating systems

IN CLASS → read, inspect, and filter the prepared result

```
taxonomy <- read.delim(  
  "data/taxonomy_final_ASV.tsv",  
  stringsAsFactors = FALSE  
)
```

```
# Preview the table  
head(taxonomy)  
names(taxonomy)
```

```
# How many ASVs?  
nrow(taxonomy)
```

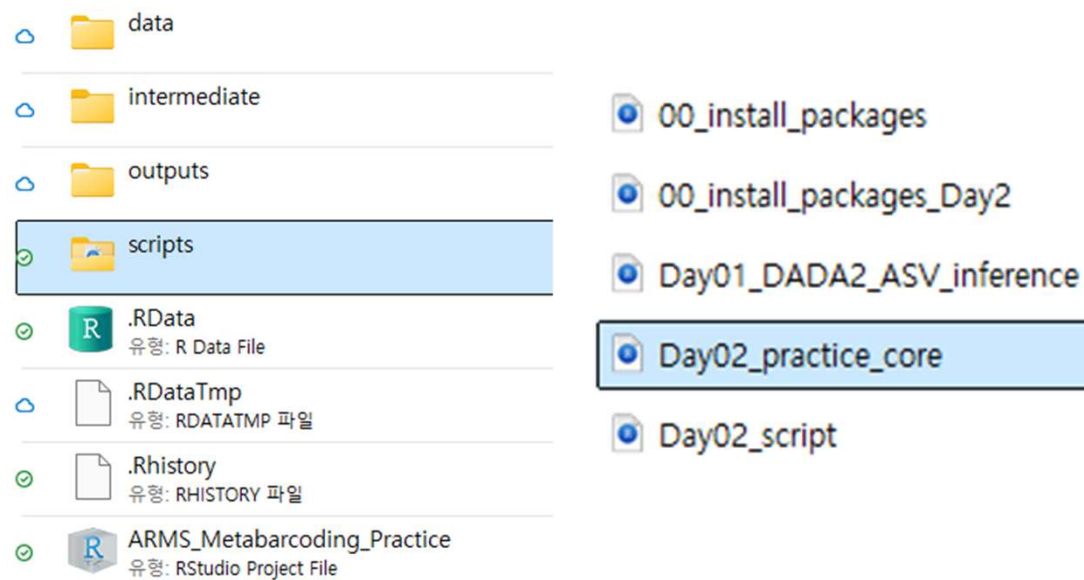
```
# Summary of BLAST assignment  
table(taxonomy$BLAST_status)
```

Open the Day 2 script

PRACTICAL

GOAL

1. Open the **Day 2 R project**.
2. Load the "Day2_practice_core.R" script.
3. Confirm the working folder.



GOAL · Run the project from its root folder and use the short script for live practice.

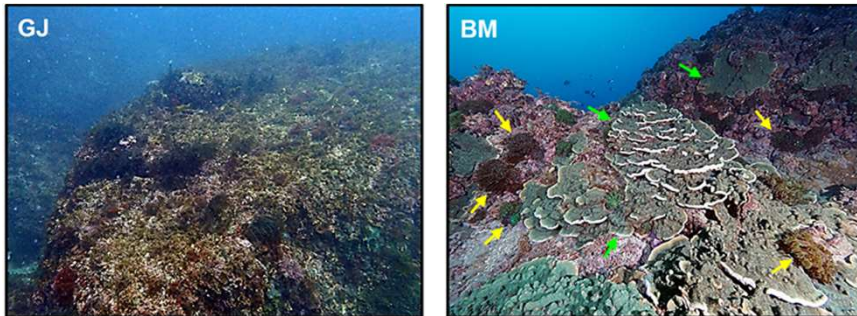
Ecological filtering

A valid BLAST match is not automatically part of the sessile-benthic target community.

KEEP

Sessile-benthic targets

- Accepted ASVs that represent the benthic community targeted in this workshop.
- E.g., Bryozoa, Cnidaria, Porifera, Mollusca, CCA



322 Accepted → 190 Included ASVs

EXCLUDE

- **Contaminants:** Bacteria · fungi · human DNA · other obvious contaminants
- **Non-target organisms:** Fish and other mobile/non-sessile taxa
- **Removed at BLAST filtering:**
 - **No_Hit** — no usable reference match
 - **Failed_cutoff** — insufficient identity and/or coverage

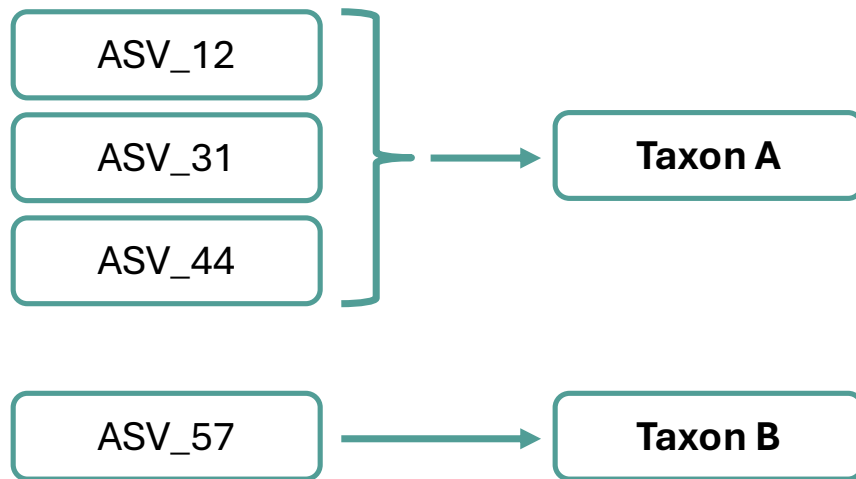
Starting dataset: 1,052 ASVs

- **730 removed during BLAST filtering**
→ *Failed cutoff or No hit*
- **322 passed BLAST filtering**
→ 132 contaminants / non-targets
→ 190 retained for ecological analysis

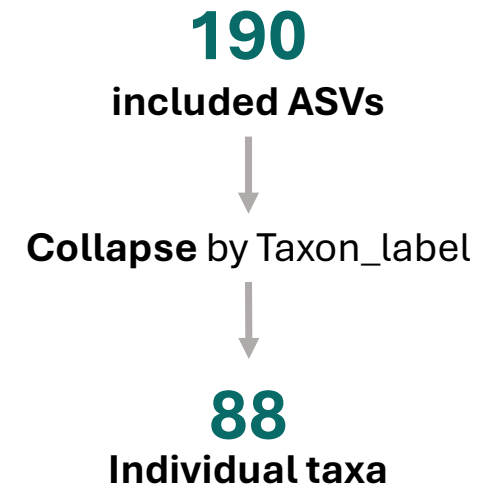
ASVs are not taxa

Multiple sequence variants can share the same taxonomic label.

Example of Taxon collapse



Collapse by taxon label



Practical 2 & 3 — Build the community matrix

PRACTICAL

GOAL

- Retain sessile-benthic target ASVs.
- Collapse ASVs sharing the same taxon label.
- Build a sample × taxon community matrix.

Expected size:

8 samples × 88 taxa.

```
# 2. Ecological filtering -----
# target ASV 선정
keep <- taxonomy$BLAST_status == "Accepted" &
  taxonomy$Status == "Include"
summary(keep)

# extract target taxa and table
target_taxonomy <- taxonomy[keep, ]
target_counts <- asv[
  match(target_taxonomy$ASV_ID, asv$ASV_ID),
  sample_names]

nrow(target_taxonomy)      # 190 ASVs

# 3. Collapse ASVs into taxa -----
taxon_counts <- rowsum(
  as.matrix(target_counts),
  target_taxonomy$Taxon_label)

community <- t(taxon_counts)
dim(community)             # 8 samples x 88 taxa
```

GOAL ·

The final community matrix contains 8 samples × 88 individual taxa.

The community matrix

Rows are samples; columns are the individual ecological taxa.

Community matrix

- Rows are samples
- Columns are the 88 individual taxa
- Values are read counts for each taxon in each sample

	<i>Clytia</i> sp.	<i>Halecium</i> sp.	<i>Laurencia</i> sp.	<i>Peyssonnelia</i> sp.	
Gangjeong1	1569	4304	0	4805	
Gangjeong2	1552	2398	0	2245	
Bomok1	268	0	10	273	
....					



10-minute break

Keep the Day R project open.

SO FAR

- Taxonomic assignment
- BLAST filtering
- Ecological filtering
- Community matrix building

NEXT

- Rarefaction
- Alpha diversity
- Community composition
- Bray–Curtis and ordination
- Community statistics
- Indicator taxa

Two kinds of diversity

Within-sample diversity and between-sample differences answer different questions.

α -diversity

How diverse is one sample?

Focus

- Diversity within a community

Examples

- Observed richness
- Shannon
- Simpson



One sample → one diversity value

β -diversity

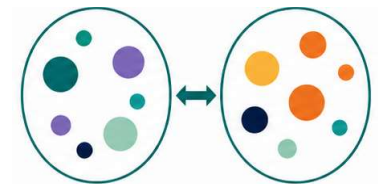
How different are sample?

Focus

- Differences between communities

Our workflow

- Bray-Curtis dissimilarity
- nMDS ordination
- ANOSIM



Sample pairs → community differences

KEY POINT ·

Alpha describes diversity within a community; beta compares communities.

Sequencing effort matters

More sampling effort can reveal more taxa even without a true biological difference.

FIELD SURVEY ANALOGY

Area A

1 survey visit



Observed:
fewer species



Area B

10 survey visit



Observed:
more species

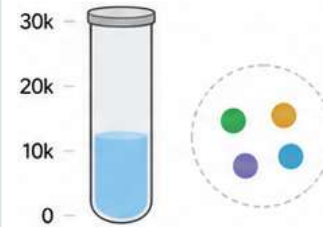


More visits = more chances
to find species

SEQUENCING ANALOGY

Sample A

8,000 reads

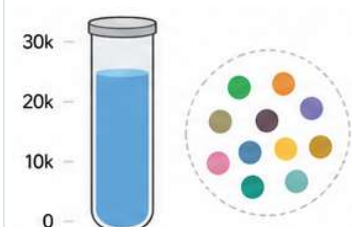


Detected taxa:
lower



Sample B

22,000 reads



Detected taxa:
higher



More reads = more chances
to detect species

KEY POINT •

Sequencing effort reflect observation effort; more reads can reveal more taxa even without a true ecological difference.

What rarefaction asks

How rapidly are new taxa recovered as reads accumulate?

TWO QUESTIONS

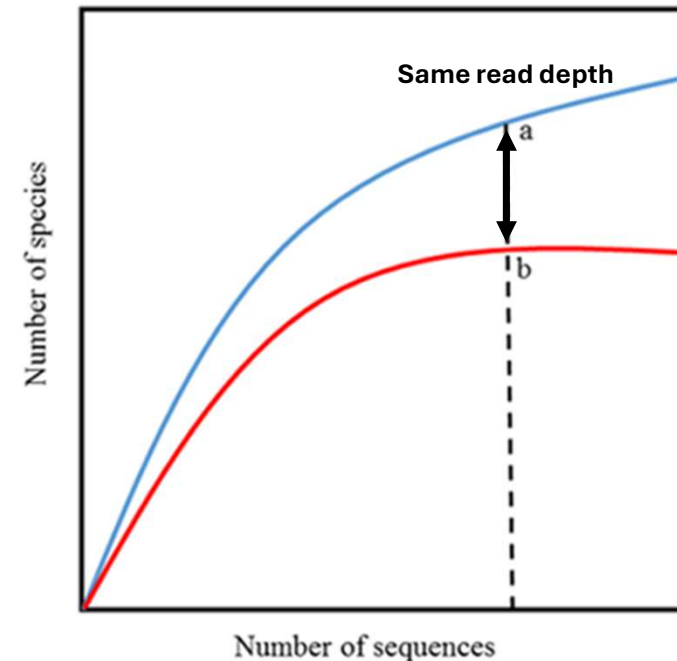
Q1: Was sequencing depth sufficient?

Curve approaches a plateau
→ Few new taxa are being recovered

Q2: Can samples be compared fairly?

Different read depths provide different opportunities to detect taxa
→ Compare samples at the **same read depth**.

Rarefaction curve



KEY POINT • Rarefaction helps evaluate sequencing sufficiency and compare richness at a common sampling effort.

Practical 4 & 5 — rarefaction

PRACTICAL

GOAL

1 · Inspect sequencing depth

Compare read depth among samples.

2 · Check richness recovery

Do the curves begin to flatten?

3 · Standardize sampling effort

Rarefy all samples to 8,499 reads.

WHAT DO YOU NOTICE?

- Read depth differs among samples.
- Most curves begin to flatten.
- The smallest library has **8,499** reads.

```
# 4. Rarefaction -----
site_colors <- c(Gangjeong = "#2C7FB8", Bomok = "#E67E22")
rarecurve(community, step = 100, label = FALSE,
          col = site_colors[metadata$Site],
          lwd = 2)
legend("bottomright", names(site_colors),
      col = site_colors, lwd = 2, bty = "n")

# 5. Rarefy to equal sequencing depth -----
sample_depth <- rowSums(community)
min_depth <- min(sample_depth)

sample_depth
min_depth

community_rare <- rrarefy(community, sample = min_depth)
rowSums(community_rare)
```

Alpha-diversity measures

Different measures capture different aspects of diversity within a sample.

HOW THE MEASURES DIFFER

Observed richness

- Number of detected taxa
- Ignores relative abundance

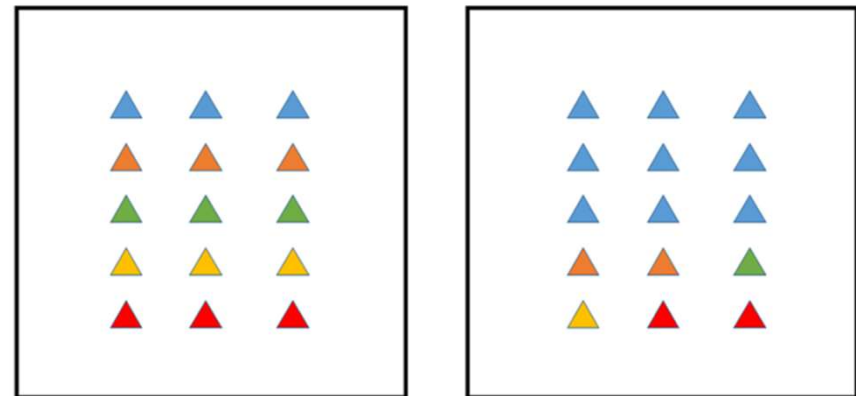
Shannon index

- Reflect richness + evenness
- Higher when taxa are more **evenly** distributed

Simpson index (D)

- Reflects richness + evenness
- More sensitive to **dominance**
- In vegan, higher value = higher diversity ($1-D$)

Example: Same richness, different evenness



Community 1

Community 2

Measure	Community 1	Community 2	Interpretation
Observed richness	5	5	Same richness
Shannon	1.61	1.18	Higher in Community 1
Simpson ($1-D$)	0.80	0.60	Higher in Community 1

KEY POINT • Richness can be identical even when Shannon and Simpson diversity differ.

Practical 6 — alpha diversity

PRACTICAL

GOAL

- Calculate Observed richness, Shannon, and Simpson indices
- Compare Gangjeong and Bomok
- Run exact Wilcoxon tests

RUN IN RSTUDIO

- Execute the code and inspect the table and boxplots.

```
# 6. Alpha diversity -----
alpha <- data.frame(
  SampleID = sample_names,
  Site = metadata$Site,
  Observed = specnumber(community_rare),
  Shannon = diversity(community_rare, "shannon"),
  Simpson = diversity(community_rare, "simpson"))
alpha

par(mfrow = c(1, 3))

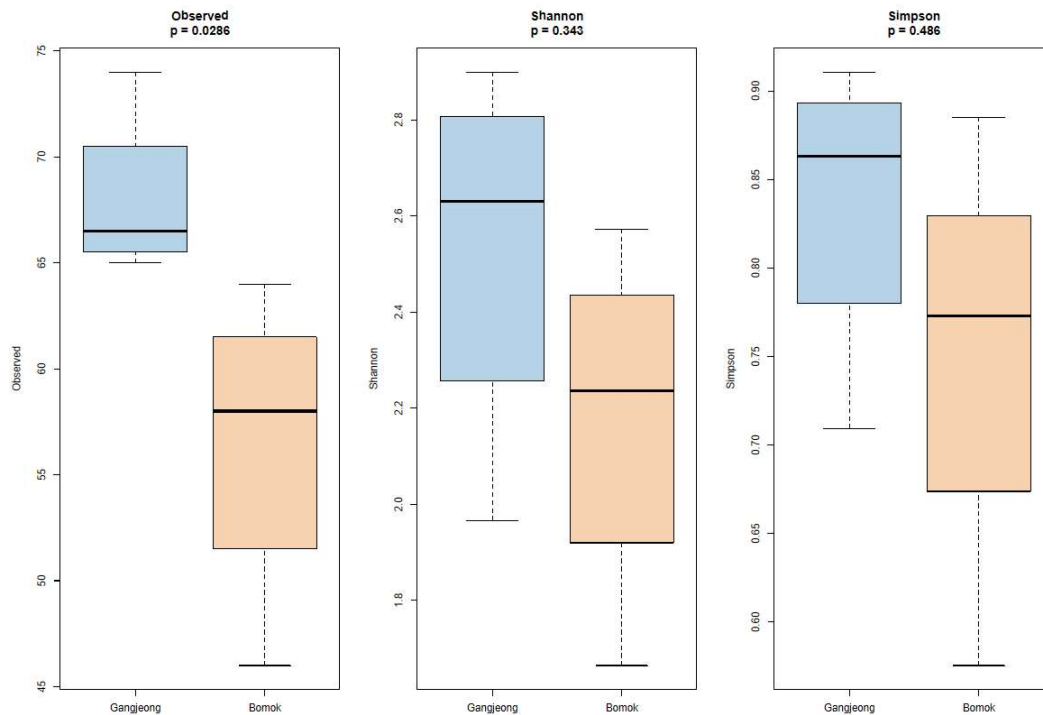
for (metric in c("Observed", "Shannon", "Simpson")) {
  test <- wilcox.test(alpha[[metric]] ~ alpha$Site, exact = TRUE)
  boxplot(alpha[[metric]] ~ alpha$Site,
    col = adjustcolor(site_colors, alpha.f = 0.35),
    main = paste0(metric, "\np = ", signif(test$p.value, 3)),
    ylab = metric)
  stripchart(alpha[[metric]] ~ alpha$Site,
    add = TRUE,
    method = "jitter",
    pch = 19,
    col = "grey25")
}
par(mfrow = c(1, 1))
```

GOAL ·

Generate the alpha-diversity results for all eight samples.

What do the alpha-diversity results show?

Only observed richness differed significantly between the two sites.



RESULTS SUMMARY

Observed richness

- GJ > BM (68.0 vs. 56.5)
- p -value = 0.0286

Shannon diversity

- GJ $\approx$ BM (2.822 vs. 2.569)
- p -value = 0.343

Simpson diversity (1-D)

- GJ $\approx$ BM (0.863 vs. 0.818)
- p -value = 0.486

WHAT DOES THIS MEAN?

- Gangjeong contained **more detected taxa**, but overall evenness and dominance structure did not show significant site differences.

KEY POINT • Gangjeong had higher observed richness, while Shannon and Simpson diversity did not differ significantly.

Practical 7 — Community composition

PRACTICAL

GOAL

1 · Aggregate taxa into broad groups

Combine the 88 taxa by Group.

2 · Convert counts to proportions

Calculate relative read proportion within each sample.

3 · Visualize community composition

Draw a stacked bar plot for all eight samples.

WHAT SHOULD WE LOOK FOR?

- Which groups dominate each sample?
- Do Gangjeong and Bomok show different composition patterns?
- Are samples within the same site similar?

```
# 7. Community composition -----
taxon_group <- target_taxonomy$Group[
  match(rownames(taxon_counts), target_taxonomy$Taxon_label)]

composition_counts <- rowsum(taxon_counts, taxon_group, reorder =
FALSE)

composition_relative <- sweep(composition_counts, 2,
                             colSums(composition_counts), "/")

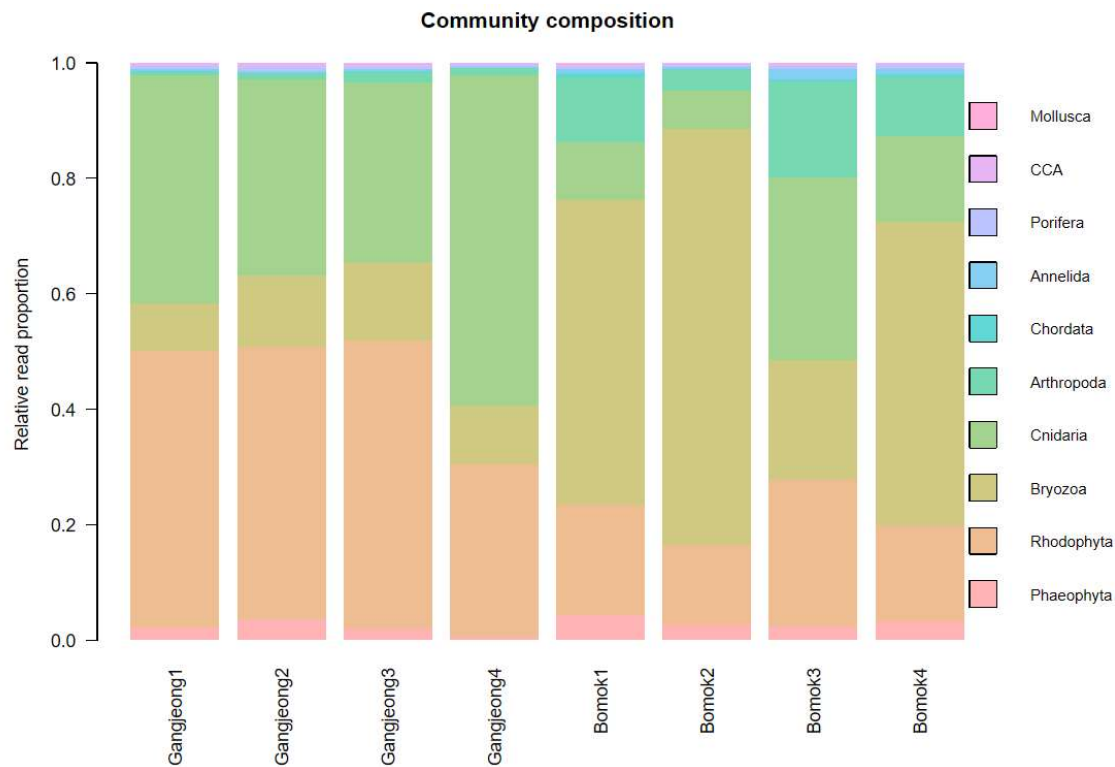
# Plotting
composition_colors <- hcl.colors(nrow(composition_relative), "Set
3")
par(mar = c(9, 5, 4, 10), xpd = TRUE)
barplot(composition_relative, col = composition_colors, border =
NA, las = 2,
        ylab = "Relative read proportion",
        main = "Community composition")
legend("topright", inset = c(-0.28, 0),
       legend = rev(rownames(composition_relative)),
       fill = rev(composition_colors),
       bty = "n", cex = 0.85)
```

GOAL ·

Visualize which broad biological groups make up each sample.

What do the composition results show?

Broad-group composition differed visually between the two sites.



RESULTS SUMMARY

Gangjeong

- Mainly composed of **Rhodophyta** and **Cnidaria**
- **Gangjeong4** shows especially high **Cnidaria**

Bomok

- Several samples are strongly dominated by **Bryozoa**
- Bomok3 shows a more mixed composition

Across samples

- Composition varies within each site
- Broad-group composition suggests **site-level differences**

WHAT DOES THIS MEAN?

- Gangjeong is largely characterized by Rhodophyta and Cnidaria.
- Bomok often contains a higher proportion of Bryozoa.
- However, samples within the same site are not identical.

KEY POINT • Broad composition suggests site-level differences, but sample-to-sample variation is also evident.

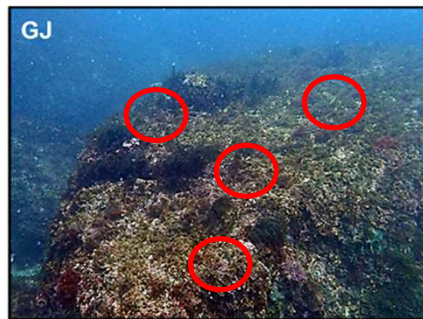
What beta diversity asks?

Do samples contain similar taxa in similar proportions?

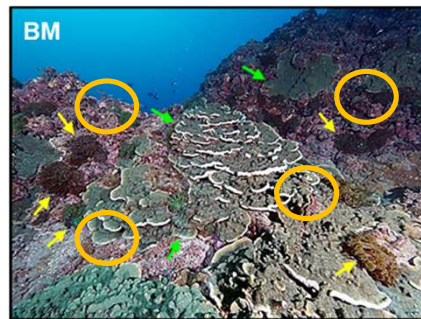
WITHIN EACH SITE

Are samples from the same site similar?

- Do they contain similar taxa?
- Are those taxa present in similar proportions?



GJ-1 GJ-2 GJ-3 GJ-4



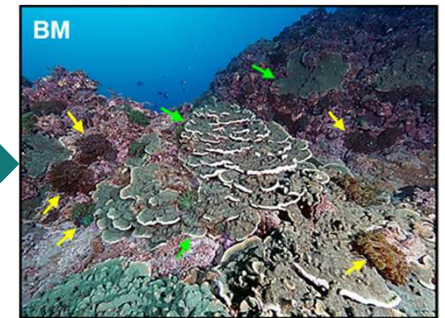
BM-1 BM-2 BM-3 BM-4

Within-site similarity

BETWEEN SITES

Are Gangjeong and Bomok different?

- Do they contain different taxa?
- Do the relative abundances of shared taxa differ?



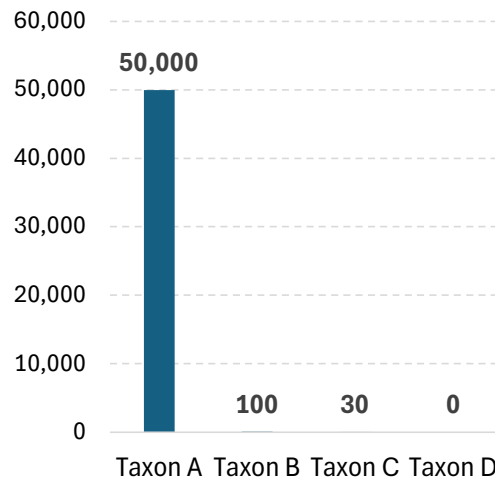
Between-site difference

KEY POINT • Beta diversity compares differences in taxon membership and community composition among samples.

Why use log1p?

Large read counts can dominate abundance-based distances.

RAW COUNTS

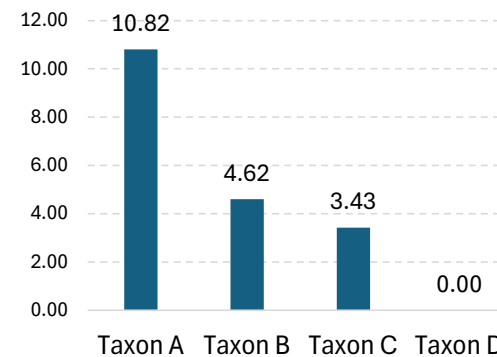


Raw read counts	
Taxon A	50,000
Taxon B	100
Taxon C	30
Taxon D	0

- Taxon/sample read counts can differ by orders of magnitude
- A few very abundant taxa may dominate Bray-Curtis distances

AFTER log1p TRANSFORMATION

$$\log_1 p(x) = \log_e(x + 1)$$



Taxon	Raw counts	log1p(x)
A	50,000	10.82
B	100	4.62
C	30	3.43
D	0	0.00

- Compress large count differences
- Reduces dominance of extremely abundant taxa

Why '+1'?

- $\log(0)$ is undefined
- Zero counts remain zero
- $0 \rightarrow \log(1) = 0$

KEY POINT • **log1p()** reduces extreme count differences while preserving zeros before Bray–Curtis analysis.

What is Bray–Curtis dissimilarity?

A single number summarizes how different two communities are.

PAIRWISE COMPARISON (Example)

Sample A (Site 1)			Sample B (Site 2)	
Taxa	Abundance		Taxa	Abundance
● Taxon 1	40	↔ Compare the abundance of all taxa	● Taxon 1	35
● Taxon 2	30		● Taxon 2	5
● Taxon 3	10		● Taxon 3	25
Total abundance (S_A) = 80			Total abundance (S_B) = 65	

HOW BRAY-CURTIS IS CALCULATED

Taxon	A	B	min(A, B)
Taxon 1	40	35	35
Taxon 2	30	5	5
Taxon 3	10	25	10
Total	80	65	$C_{AB} = \sum \min(A, B) = 50$

Bray-Curtis formula

$$BC_{AB} = 1 - \frac{2C_{AB}}{S_A + S_B}$$

Plug in the values

$$BC_{AB} = 1 - \frac{2(50)}{80 + 65} = 1 - \frac{100}{145} = 0.31$$

BRAY-CURTIS DISSIMILARITY

Often referred to as a community distance

0 ← Similar → 1
different

- Use abundance information
- Calculated for every pair of samples
- Produces a dissimilarity matrix

Interpretation

- 0 → identical composition
- $0 < BC < 1$ → more different
- 1 → completely different

Distance (dissimilarity) matrix

	A	B	C
A	0	0.42	0.67
B	0.42	0	0.35
C	0.67	0.35	0

KEY POINT • Bray–Curtis converts differences in taxon abundance into pairwise community dissimilarities.

nMDS: Visualizing community dissimilarities

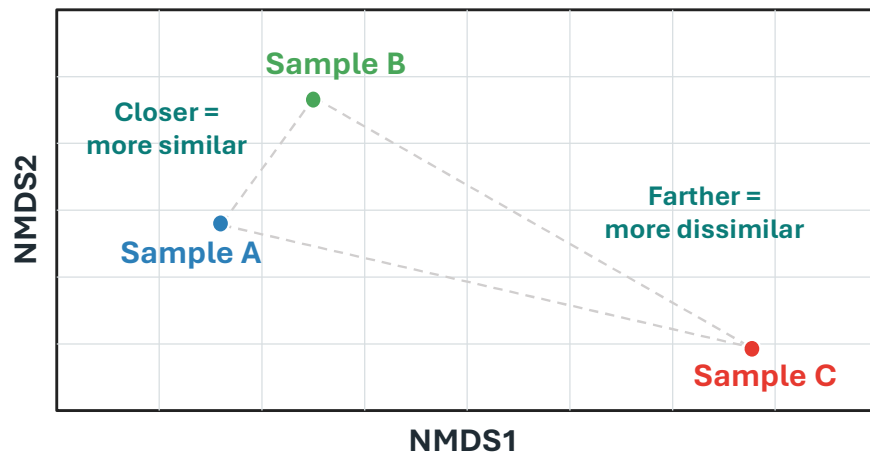
nMDS turns pairwise community dissimilarities into a 2D map while preserving their rank order

WHAT nMDS DOES

Bray–Curtis
dissimilarities



nMDS
(2 dimensions)

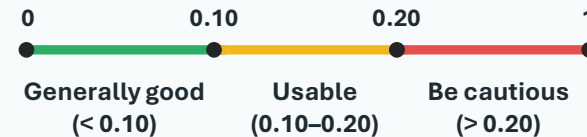


- Each point = one sample (community)
- NMDS1 and NMDS2 have no direct ecological meaning

STRESS: HOW GOOD IS THE FIT?

Stress tells us how well the 2D map represents the original dissimilarity relationships.

Stress value (lower is better)



- Lower stress = better 2D representation
- Always check stress before interpretation
- 2D distances are not exact Bray–Curtis values.

KEY POINT • Closer points represent more similar communities, and lower stress means a more reliable 2D representation.

Practical 8 — dissimilarity and nMDS

PRACTICAL

GOAL

1 • Transform the community matrix

Apply `log1p()` to reduce extreme count differences.

2 • Calculate Bray–Curtis dissimilarity

Compare every pair of samples.

3 • Generate an nMDS ordination

Visualize community relationships in 2D.

4 • Check the stress value

Evaluate how well the ordination represents the dissimilarities.

WHAT SHOULD WE CHECK?

- Do Gangjeong and Bomok separate visually?
- Are samples within each site relatively close?
- Is the stress value sufficiently low?

```
# 8. Bray-Curtis and nMDS -----

community_log <- log1p(community)

bray <- vegdist(community_log, method = "bray")
print(bray)

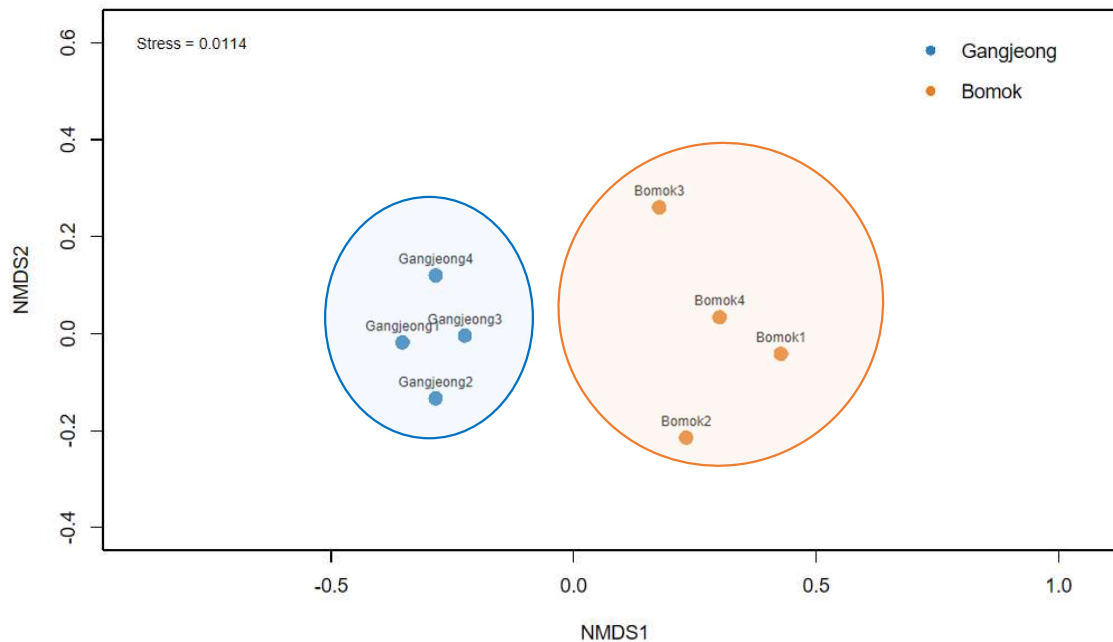
nmds <- metaMDS(community_log, distance = "bray", k = 2, trymax = 100,
               autotransform = FALSE, trace = FALSE)
nmds$stress

plot(nmds, type = "n")
points(nmds, display = "sites", pch = 19, cex = 1.5,
       col = site_colors[metadata$Site])
text(nmds, display = "sites", labels = sample_names, pos = 3, cex = 0.7)
legend("topright", names(site_colors), col = site_colors, pch = 19, bty = "n")
```

GOAL • Calculate and visualize community differences among the eight samples.

What do the nMDS results show?

Gangjeong and Bomok show clear visual separation in community composition.



RESULTS SUMMARY

Gangjeong

- Samples form a relatively tight cluster
- Located together in nMDS space

Bomok

- Samples are more dispersed than Gangjeong
- Clearly separated from Gangjeong along the main ordination pattern

Stress

- Stress = 0.0114
- Very low distortion in the 2D representation

KEY POINT • Gangjeong and Bomok appear compositionally distinct, but statistical testing is still required.

ANOSIM (Analysis of Similarity)

ANOSIM tests whether communities from different groups are more dissimilar than communities within the same group.

1 • IDEA

Compare rank similarities between and within groups.

Within-group

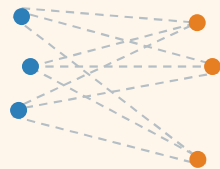
pairs from the same site



Average rank
dissimilarity = r_W

Between-group

pairs from different sites



Average rank
dissimilarity = r_B



If communities differ by site, between-group similarities (r_B) will be lower than within-group similarities (r_W).

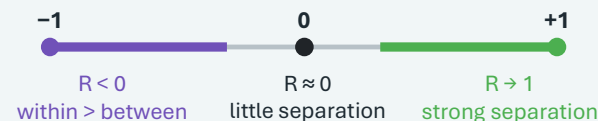
2 • TEST STATISTIC R

ANOSIM uses the ranks of all pairwise dissimilarities.

$$R = \frac{r_B - r_W}{M/2}$$

- r_B : mean rank of between-site dissimilarities
- r_W : mean rank of within-site dissimilarities
- M = total number of pairwise dissimilarities

INTERPRETATION OF R



3 • HOW IT WORKS

1 Input

Bray–Curtis dissimilarity matrix and group labels (e.g., sites).

2 Rank dissimilarities

Order all sample-pair dissimilarities

3 Compare mean ranks

Average ranks for
Between-site ranks vs. within-site ranks

4 Compute R statistics

Summarize the separation strength

5 Assess significance

Use permutations of group labels (default 999) to obtain a p -value

- R = separation strength
- p -value = permutation support

KEY POINT • Higher R means stronger group separation; the p -value tells us whether it is statistically supported.

Practical 9 — test site effects (ANOSIM)

PRACTICAL

GOAL

1 • Run ANOSIM

Test the difference between Gangjeong and Bomok.

2 • Check the R statistic

Higher R = stronger group separation.

3 • Check the p-value

Does the permutation test support the difference?

WHAT DO YOU NOTICE?

- R = 0.927
- p-value = 0.026
- ➔ Strong separation between the two sites

9. ANOSIM -----

```
anosim_result <- anosim( bray, metadata$Site, permutations = 999)
```

```
anosim_result
```

Result

```
> anosim_result
```

Call:

```
anosim(x = bray, grouping = metadata$Site, permutations = 999)
```

Dissimilarity: bray

ANOSIM statistic R: 0.9792 → **separation strength**

Significance: 0.026 → **statistical support**

Permutation: free

Number of permutations: 999

Which taxa characterize a site?

Indicator analysis (IndVal) connects community-level differences to individual taxa.

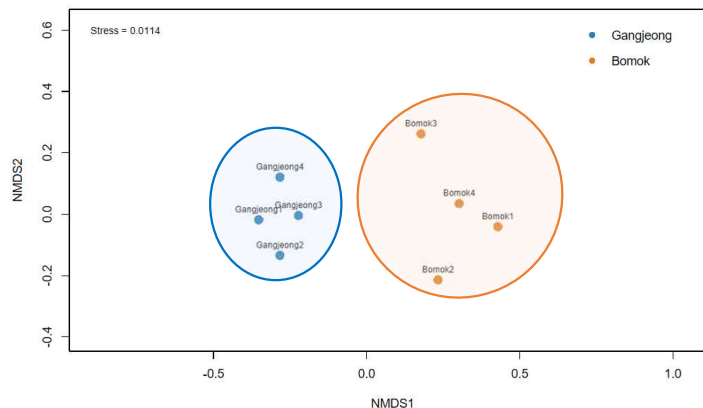
Community –level result

Gangjeong and Bomok differ in overall community composition.

ANOSIM

→ $R = 0.979$

→ $p = 0.026$



The sites are different
— but which taxa drive this pattern?

Taxon-level question

Which taxa characterize each site?

Indicator analysis (IndVal) looks for taxa that are:

- Strongly associated with one site
- Consistently detected within that site

These two properties are called **specificity** and **fidelity**.

IndVal: Specificity and Fidelity

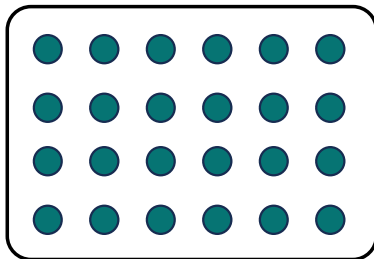
A useful indicator is both concentrated in one site and consistently present.

$$\text{IndVal} = \sqrt{A \times B}$$

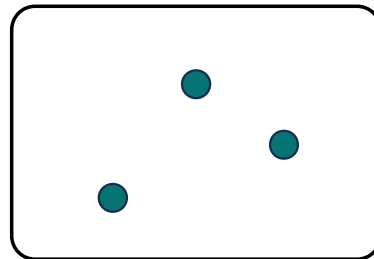
SPECIFICITY (A)

Is this taxon mainly associated with one site?

Site 1



Site 2



● : taxon 1

High when most abundance is concentrated in one site.

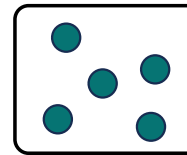
$$A = \frac{\text{mean abundance in target site}}{\text{sum of mean abundance across all sites}}$$

FIDELITY (B)

Does the taxon occur consistently within that site?

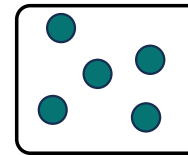
For site 1:

Sample 1



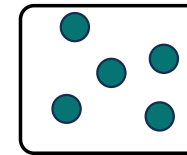
✓

Sample 2



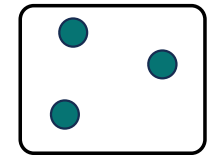
✓

Sample 3



✓

Sample 4



✓

High when the taxon is present in most samples of that site.

$$B = \frac{\text{number of samples where present}}{\text{total samples in the site}}$$

Practical 10 — indicator taxa (IndVal)

PRACTICAL

GOAL

1 · Calculate IndVal

Evaluate all taxa for each site.

2 · Inspect A and B

- A = specificity
- B = fidelity

3 · Identify indicator taxa

Look for high IndVal and $p < 0.05$.

WHAT SHOULD WE CHECK?

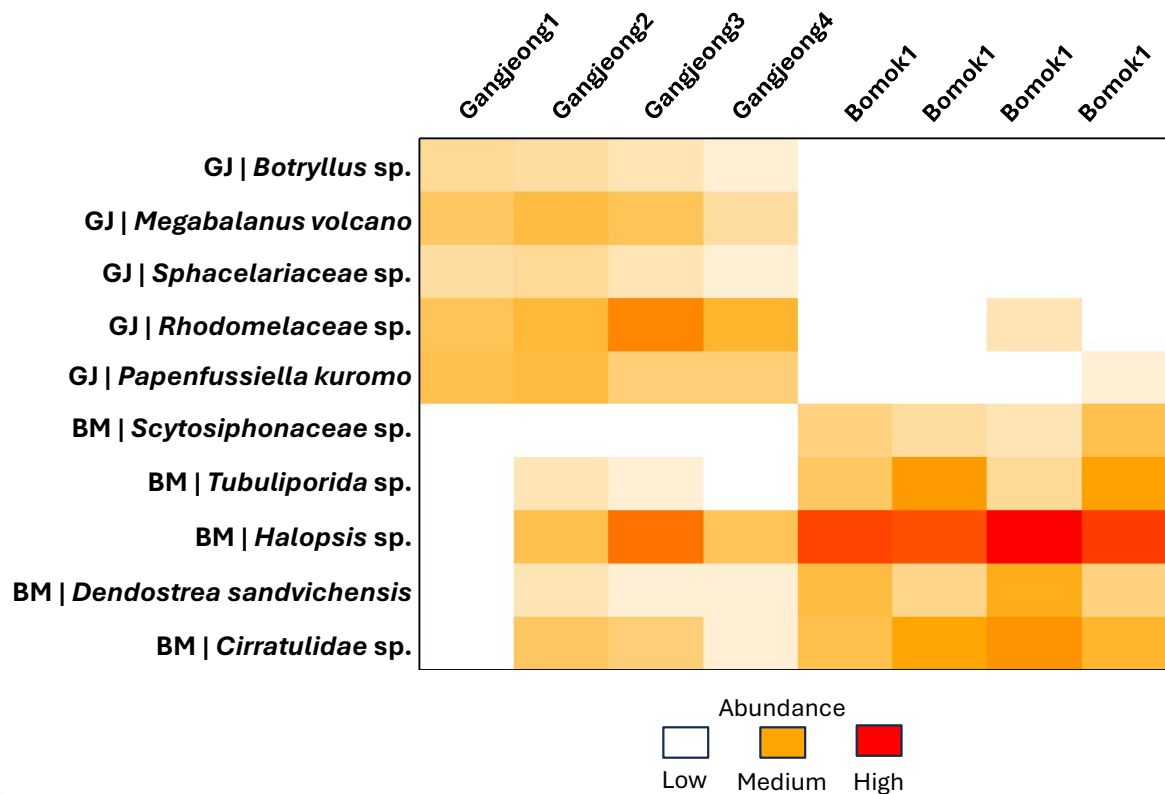
- Which site is the taxon associated with?
- Are A and B both high?
- Is the permutation p-value < 0.05 ?

```
# 10. Indicator taxa -----  
set.seed(20260821)  
  
indval <- multipatt(  
  community,  
  metadata$Site,  
  func = "IndVal.g",  
  control = permute::how(nperm = 999))  
  
summary(indval, alpha = 0.05, indvalcomp = TRUE)
```

GOAL · IndVal identifies taxa that are strongly and consistently associated with a site.

What do the indicator-taxon results show?

IndVal links site-level community differences to specific taxa and their broad taxonomic groups.



RESULTS SUMMARY

IndVal pattern

- Gangjeong and Bomok each have site-specific indicator taxa
- Heatmap matches the nMDS/ANOSIM pattern
- Indicator taxa are abundant mainly in their associated site

Site	Indicator taxon	Group
Gangjeong	<i>Botryllus</i> sp.	Chordata
	<i>Megabalanus volcano</i>	Arthropoda
	<i>Sphacelariaceae</i> sp.	Phaeophyta
	<i>Rhodomelaceae</i> sp.	Rhodophyta
	<i>Papenfussiella kuromo</i>	Phaeophyta
Bomok	<i>Scytosiphonaceae</i> sp.	Phaeophyta
	<i>Tubuliporida</i> sp.	Bryozoa
	<i>Halopsis</i> sp.	Cnidaria
	<i>Dendostrea sandwichensis</i>	Mollusca
	<i>Cirratulidae</i> sp.	Annelida

KEY POINT · Indicator taxa identify which taxa best characterize Gangjeong and Bomok.

Today's four questions — answered

From ASVs to ecological interpretation.

1. Taxonomy & filtering

What organisms matched our ASVs, and which did we analyze?

- ✓ **322 ASVs** passed BLAST filtering
- ✓ **190 ASVs** retained as target sessile benthic organisms
- ✓ Collapsed into **88 taxa**

Download GenBank Graphics

Balanus trigonus voucher CR7 cytochrome c oxidase subunit I (COX1) gene, partial cds; Sequence ID: PP718007.1 Length: 658 Number of Matches: 1

Range: 11 to 276 GenBank Graphics

Score	Expect	Identical	Coverage	Strand
5150 bits (276)	4e-540	276/276 (100%)	0/276 (0%)	Plus/Minus

Query 1 AACCTTATATTTAAATTTTGGGGGCTGATCAGCTATAGGAGACAGCTTAAAGATGTT 60
 Sbjct 1 AACCTTATATTTAAATTTTGGGGGCTGATCAGCTATAGGAGACAGCTTAAAGATGTT 60

Query 61 AATTGAGAGCTGAATAGGTGACACAGAGAGTTTAAATGGTGATGACAAATTTATAATGT 120
 Sbjct 61 AATTGAGAGCTGAATAGGTGACACAGAGAGTTTAAATGGTGATGACAAATTTATAATGT 120

Query 121 TATTGTTACAGCTCAAGCTTTTATATAATTTTCTTATAGTAAATAGCTATTATAATTTG 180
 Sbjct 121 TATTGTTACAGCTCAAGCTTTTATATAATTTTCTTATAGTAAATAGCTATTATAATTTG 180

Query 181 GGATTTGGAAATGACCTTCTTCTTAACTAGTGGGCGCTGATATGGCATTCCTCG 240
 Sbjct 181 GGATTTGGAAATGACCTTCTTCTTAACTAGTGGGCGCTGATATGGCATTCCTCG 240

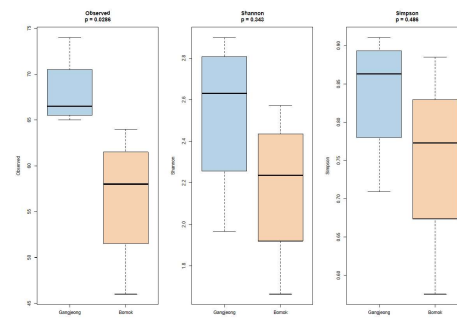
Query 241 ATTAATAATAGAGATTTTGAATCTTCTTCTTCTTCTTCTTCTTCTTCTTCTTCTTCTT 276
 Sbjct 241 ATTAATAATAGAGATTTTGAATCTTCTTCTTCTTCTTCTTCTTCTTCTTCTTCTTCTT 276

**BLAST + ecological filtering
+ taxonomic collapse**

2. α -diversity

How diverse is each community?

- ✓ Observed richness: GJ > BM (68.0 vs. 56.5, $p = 0.0286$)
- ✓ Shannon and Simpson: **no significant difference**

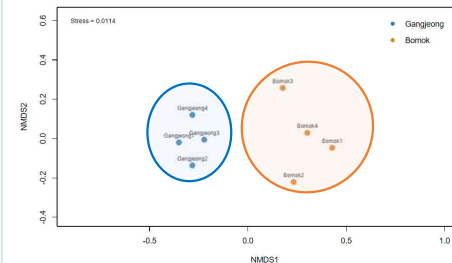


**Rarefaction
+ richness / Shannon / Simpson**

3. β -diversity

How do the two communities differ?

- ✓ nMDS showed clear separation by site
- ✓ ANOSIM: **$R = 0.979$, $p = 0.026$**
→ Significant site-level difference

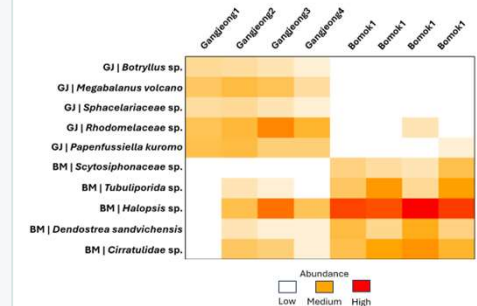


**$\log_1 p$ + Bray-Curtis
+ nMDS + ANOSIM**

4. Indicator taxa

Which taxa characterize each site?

- ✓ **Gangjeong:** Mainly algae, with sessile invertebrates also represented
- ✓ **Bomok:** Spanning a broader range of benthic groups



IndVal + indicator-taxon heatmap

KEY POINT ·

Each step answers a different question — from taxonomic identity to community ecology.

- The end -

Thank you!

